

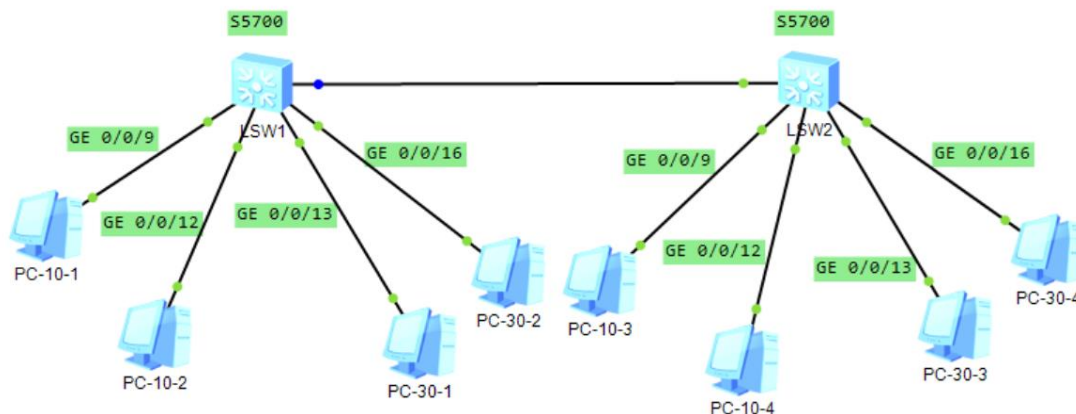
Lab-3 inter-VLAN Routing

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一、Lab-3-1: VLANIF Configuration

Q3-1.1~3-1.8:

(1) 网络拓补结构图如下：



(2) 交换机 LSW1 的 IP 路由表如下：

```
[LSW1]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 6           Routes : 6

Destination/Mask    Proto   Pre  Cost           Flags NextHop         Interface
-----
127.0.0.0/8         Direct  0    0               D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct  0    0               D    127.0.0.1         InLoopBack0
192.168.10.0/24      Direct  0    0               D    192.168.10.1      Vlanif10
192.168.10.1/32      Direct  0    0               D    127.0.0.1         Vlanif10
192.168.30.0/24      Direct  0    0               D    192.168.30.1      Vlanif30
192.168.30.1/32      Direct  0    0               D    127.0.0.1         Vlanif30
```

(3) PC-10-1 和 PC-10-3 可以 ping 通，效果如下：

```
PC>ping 192.168.10.13

Ping 192.168.10.13: 32 data bytes, Press Ctrl_C to break
From 192.168.10.13: bytes=32 seq=1 ttl=128 time=78 ms
From 192.168.10.13: bytes=32 seq=2 ttl=128 time=109 ms
From 192.168.10.13: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.10.13: bytes=32 seq=4 ttl=128 time=93 ms
From 192.168.10.13: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.10.13 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/74/109 ms
```

(4) PC-30-1 和 PC-30-3 可以 ping 通，效果如下：

```
PC>ping 192.168.30.13

Ping 192.168.30.13: 32 data bytes, Press Ctrl_C to break
From 192.168.30.13: bytes=32 seq=1 ttl=128 time=78 ms
From 192.168.30.13: bytes=32 seq=2 ttl=128 time=78 ms
From 192.168.30.13: bytes=32 seq=3 ttl=128 time=109 ms
From 192.168.30.13: bytes=32 seq=4 ttl=128 time=94 ms
From 192.168.30.13: bytes=32 seq=5 ttl=128 time=78 ms

--- 192.168.30.13 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 78/87/109 ms
```

(5) PC-10-1 和 PC-30-3 可以 ping 通，效果如下：

```
PC>ping 192.168.30.13

Ping 192.168.30.13: 32 data bytes, Press Ctrl_C to break
From 192.168.30.13: bytes=32 seq=1 ttl=127 time=203 ms
From 192.168.30.13: bytes=32 seq=2 ttl=127 time=94 ms
From 192.168.30.13: bytes=32 seq=3 ttl=127 time=47 ms
From 192.168.30.13: bytes=32 seq=4 ttl=127 time=78 ms
From 192.168.30.13: bytes=32 seq=5 ttl=127 time=94 ms

--- 192.168.30.13 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/103/203 ms
```

(6) “tracert 192.168.30.13” 的结果如下：

```
PC>tracert 192.168.30.13

tracert to 192.168.30.13, 8 hops max
(ICMP), press Ctrl+C to stop
 1  192.168.10.1    15 ms  32 ms  31 ms
 2  192.168.30.13   94 ms  109 ms  78 ms
```

(7) 对于 PC-10-1 和 PC-10-3 之间捕获的结果如下，进出时捕获的 VLAN ID 都是 10：

No.	Time	Source	Destination	Protocol	Length	Info
38	76.953000	192.168.10.11	192.168.10.13	ICMP	78	Echo (ping) request id=0x6eed, seq=1/256, ttl=128 (reply in 39)
39	77.000000	192.168.10.13	192.168.10.11	ICMP	78	Echo (ping) reply id=0x6eed, seq=1/256, ttl=128 (request in 38)
40	77.266000	HuaweiTechno_3d:61::	Spanning-tree-(for-...	STP	119	MST. Root = 32768/0/4c:1f:cc:3d:61:d4 Cost = 0 Port = 0x8018
41	78.031000	192.168.10.11	192.168.10.13	ICMP	78	Echo (ping) request id=0x6fed, seq=2/512, ttl=128 (reply in 42)
42	78.078000	192.168.10.13	192.168.10.11	ICMP	78	Echo (ping) reply id=0x6fed, seq=2/512, ttl=128 (request in 41)
43	79.125000	192.168.10.11	192.168.10.13	ICMP	78	Echo (ping) request id=0x70ed, seq=3/768, ttl=128 (reply in 44)
44	79.156000	192.168.10.13	192.168.10.11	ICMP	78	Echo (ping) reply id=0x70ed, seq=3/768, ttl=128 (request in 43)
45	79.500000	HuaweiTechno_3d:61::	Spanning-tree-(for-...	STP	119	MST. Root = 32768/0/4c:1f:cc:3d:61:d4 Cost = 0 Port = 0x8018
46	80.203000	192.168.10.11	192.168.10.13	ICMP	78	Echo (ping) request id=0x71ed, seq=4/1024, ttl=128 (reply in 47)
47	80.234000	192.168.10.13	192.168.10.11	ICMP	78	Echo (ping) reply id=0x71ed, seq=4/1024, ttl=128 (request in 46)
48	81.281000	192.168.10.11	192.168.10.13	ICMP	78	Echo (ping) request id=0x72ed, seq=5/1280, ttl=128 (reply in 49)

> Frame 41: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface	0000	54 89 98 9d 2e fc 54 89 98 ed 3a f4 81 00 00 0a	T.....T.....
> Ethernet II, Src: HuaweiTechno_ed:3a:f4 (54:89:98:ed:3a:f4), Dst: HuaweiTechno	0010	08 00 45 00 00 3c ed 6f 40 00 80 01 77 e8 c0 a8	..E...<O@...W...
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 10	0020	0a 0b c0 a8 0a 0d 08 00 16 8f 6f ed 00 02 08 09	O.....
> Internet Protocol Version 4, Src: 192.168.10.11, Dst: 192.168.10.13	0030	0a 0b 0c 0d 0e 0f 10 11 12 13 14 15 16 17 18 19	
> Internet Control Message Protocol	0040	1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27	 '#%&'

38	76.953000	192.168.10.11	192.168.10.13	ICMP	78 Echo (ping) request	id=0x6eed, seq=1/256, ttl=128 (reply in 39)	
39	77.000000	192.168.10.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0x6eed, seq=1/256, ttl=128 (request in 38)	
40	77.266000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	
41	78.031000	192.168.10.11	192.168.10.13	ICMP	78 Echo (ping) request	id=0x6fed, seq=2/512, ttl=128 (reply in 42)	
42	78.078000	192.168.10.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0x6fed, seq=2/512, ttl=128 (request in 41)	
43	79.125000	192.168.10.11	192.168.10.13	ICMP	78 Echo (ping) request	id=0x70ed, seq=3/768, ttl=128 (reply in 44)	
44	79.156000	192.168.10.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0x70ed, seq=3/768, ttl=128 (request in 43)	
45	79.500000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	
46	80.203000	192.168.10.11	192.168.10.13	ICMP	78 Echo (ping) request	id=0x71ed, seq=4/1024, ttl=128 (reply in 47)	
47	80.234000	192.168.10.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0x71ed, seq=4/1024, ttl=128 (request in 46)	
48	81.281000	192.168.10.11	192.168.10.13	ICMP	78 Echo (ping) request	id=0x72ed, seq=5/1280, ttl=128 (reply in 49)	

> Frame 42: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interfa				0000	54 89 98 ed 3a f4 54 89 98 9d 2e fc 81 00 00 0a	T...T... ..
> Ethernet II, Src: HuaweiTechno_9d:2e:fc (54:89:98:9d:2e:fc), Dst: HuaweiTechno				0010	08 00 45 00 00 3c ed 6f 40 00 80 01 77 a8 c0 a8	..E-<..o @..W...
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 10				0020	0a 0d c0 a8 0a 0b 00 00 1e 8f 6f ed 00 02 08 09	o.....
> Internet Protocol Version 4, Src: 192.168.10.13, Dst: 192.168.10.11				0030	0a 0b 0c 0d 0e 0f 10 11 12 13 14 15 16 17 18 19	
> Internet Control Message Protocol				0040	1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27	 ! "##\$%

(8) 对于 PC-10-1 和 PC-30-3 之间捕获的结果如下，进出时捕获的 VLAN ID 都是 30:

8	8.484000	192.168.30.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0xdeee, seq=2/512, ttl=128 (request in 7)	
9	8.890000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	
10	9.531000	192.168.10.11	192.168.30.13	ICMP	78 Echo (ping) request	id=0xdfee, seq=3/768, ttl=127 (reply in 11)	
11	9.562000	192.168.30.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0xdfee, seq=3/768, ttl=128 (request in 10)	
12	10.625000	192.168.10.11	192.168.30.13	ICMP	78 Echo (ping) request	id=0xe0ee, seq=4/1024, ttl=127 (reply in 13)	
13	10.656000	192.168.30.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0xe0ee, seq=4/1024, ttl=128 (request in 12)	
14	11.109000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	
15	11.703000	192.168.10.11	192.168.30.13	ICMP	78 Echo (ping) request	id=0xe1ee, seq=5/1280, ttl=127 (reply in 16)	
16	11.750000	192.168.30.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0xe1ee, seq=5/1280, ttl=128 (request in 15)	
17	13.320000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	
18	15.515000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	

> Frame 10: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interfa				0000	54 89 98 c4 50 75 4c 1f cc 3d 61 d4 81 00 00 1e	T...PuL...=a....
> Ethernet II, Src: HuaweiTechno_3d:61:d4 (4c:1f:cc:3d:61:d4), Dst: HuaweiTechno				0010	08 00 45 00 00 3c ee de 40 00 7f 01 63 79 c0 a8	..E-<..@...cy...
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 30				0020	0a 0b c0 a8 1e 0d 08 00 a6 8c df ee 00 03 08 09	@...by...
> Internet Protocol Version 4, Src: 192.168.10.11, Dst: 192.168.30.13				0030	0a 0b 0c 0d 0e 0f 10 11 12 13 14 15 16 17 18 19	
> Internet Control Message Protocol				0040	1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27	 ! "##\$%

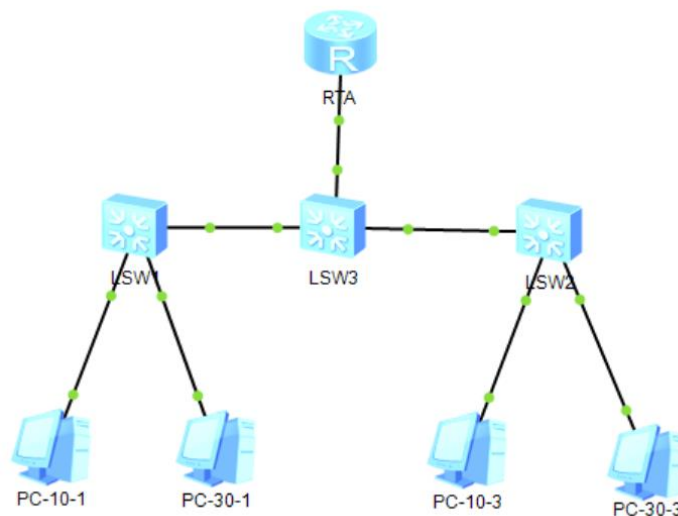
8	8.484000	192.168.30.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0xdeee, seq=2/512, ttl=128 (request in 7)	
9	8.890000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	
10	9.531000	192.168.10.11	192.168.30.13	ICMP	78 Echo (ping) request	id=0xdfee, seq=3/768, ttl=127 (reply in 11)	
11	9.562000	192.168.30.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0xdfee, seq=3/768, ttl=128 (request in 10)	
12	10.625000	192.168.10.11	192.168.30.13	ICMP	78 Echo (ping) request	id=0xe0ee, seq=4/1024, ttl=127 (reply in 13)	
13	10.656000	192.168.30.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0xe0ee, seq=4/1024, ttl=128 (request in 12)	
14	11.109000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	
15	11.703000	192.168.10.11	192.168.30.13	ICMP	78 Echo (ping) request	id=0xe1ee, seq=5/1280, ttl=127 (reply in 16)	
16	11.750000	192.168.30.13	192.168.10.11	ICMP	78 Echo (ping) reply	id=0xe1ee, seq=5/1280, ttl=128 (request in 15)	
17	13.320000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	
18	15.515000	HuaweiTechno_3d:61:...	Spanning-tree (for-...	STP	119 MST. Root = 32768/0/4c:1f:cc:3d:61:d4	Cost = 0 Port = 0x8018	

> Frame 11: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interfa				0000	4c 1f cc 3d 61 d4 54 89 98 c4 50 75 81 00 00 1e	L...aT...Pu....
> Ethernet II, Src: HuaweiTechno_c4:50:75 (54:89:98:c4:50:75), Dst: HuaweiTechno				0010	08 00 45 00 00 3c ee de 40 00 80 01 62 79 c0 a8	..E-<..@...by...
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 30				0020	1e 0d c0 a8 0a 0b 00 00 a6 8c df ee 00 03 08 09	@...by...
> Internet Protocol Version 4, Src: 192.168.30.13, Dst: 192.168.10.11				0030	0a 0b 0c 0d 0e 0f 10 11 12 13 14 15 16 17 18 19	
> Internet Control Message Protocol				0040	1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27	 ! "##\$%

二、Lab-3-2: One-armed Router Configuration

Q3-2.1~3-2.7:

(1) 网络拓补结构图如下:



(2) RTA 上的 IP 路由表如下:

```
[RTA]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 10          Routes : 10

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
      127.0.0.0/8     Direct  0     0           D  127.0.0.1         InLoopBack0
      127.0.0.1/32     Direct  0     0           D  127.0.0.1         InLoopBack0
127.255.255.255/32   Direct  0     0           D  127.0.0.1         InLoopBack0
      192.168.10.0/24  Direct  0     0           D  192.168.10.1      GigabitEthernet
0/0/2.10
      192.168.10.1/32  Direct  0     0           D  127.0.0.1         GigabitEthernet
0/0/2.10
      192.168.10.255/32 Direct  0     0           D  127.0.0.1         GigabitEthernet
0/0/2.10
      192.168.30.0/24  Direct  0     0           D  192.168.30.1      GigabitEthernet
0/0/2.30
      192.168.30.1/32  Direct  0     0           D  127.0.0.1         GigabitEthernet
0/0/2.30
      192.168.30.255/32 Direct  0     0           D  127.0.0.1         GigabitEthernet
0/0/2.30
255.255.255.255/32   Direct  0     0           D  127.0.0.1         InLoopBack0
```

(3) PC-10-1 和 PC-10-3 可以 ping 通, 结果如下:

```
PC>ping 192.168.10.13

Ping 192.168.10.13: 32 data bytes, Press Ctrl_C to break
From 192.168.10.13: bytes=32 seq=1 ttl=128 time=78 ms
From 192.168.10.13: bytes=32 seq=2 ttl=128 time=63 ms
From 192.168.10.13: bytes=32 seq=3 ttl=128 time=93 ms
From 192.168.10.13: bytes=32 seq=4 ttl=128 time=78 ms
From 192.168.10.13: bytes=32 seq=5 ttl=128 time=79 ms

--- 192.168.10.13 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 63/78/93 ms
```

(4) PC-10-1 上调用“tracert 192.168.30.13”命令效果如下:

```
PC>tracert 192.168.30.13

tracert to 192.168.30.13, 8 hops max
(ICMP), press Ctrl+C to stop
 1  192.168.10.1      62 ms  63 ms  78 ms
 2  *192.168.30.13   125 ms  125 ms
```

(5) LSW3 上接口 2 捕获到广播帧如下, 数据帧携带的 VLAN ID 为 10:

```
77 166.438000 HuaweiTechno_e9:33:.. Spanning-tree(for-.. STP 119 MST. Root = 32768/0/4c:1f:cc:41:75:f2 Cost = 20000 Port = 0x8002
78 168.641000 HuaweiTechno_e9:33:.. Spanning-tree(for-.. STP 119 MST. Root = 32768/0/4c:1f:cc:41:75:f2 Cost = 20000 Port = 0x8002
79 170.594000 HuaweiTechno_c0:47:.. Broadcast ARP 64 who has 192.168.10.13? Tell 192.168.10.11
80 170.828000 HuaweiTechno_e9:33:.. Spanning-tree(for-.. STP 119 MST. Root = 32768/0/4c:1f:cc:41:75:f2 Cost = 20000 Port = 0x8002
81 173.031000 HuaweiTechno_e9:33:.. Spanning-tree(for-.. STP 119 MST. Root = 32768/0/4c:1f:cc:41:75:f2 Cost = 20000 Port = 0x8002
82 175.266000 HuaweiTechno_e9:33:.. Spanning-tree(for-.. STP 119 MST. Root = 32768/0/4c:1f:cc:41:75:f2 Cost = 20000 Port = 0x8002

> Frame 18: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interfa
> Ethernet II, Src: HuaweiTechno_c0:47:8a (54:89:98:c0:47:8a), Dst: Broadcast (f
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 10
> Address Resolution Protocol (request)

0000 ff ff ff ff ff 54 89 98 c0 47 8a 81 00 00 0a .....T..G....
0010 08 06 00 01 08 00 06 04 00 01 54 89 98 c0 47 8a .....T..G....
0020 c0 a8 0a 0b ff ff ff ff ff ff c0 a8 0a 0d 30 0c .....G.....
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....G.....
```

No.	Time	Source	Destination	Protocol	Length	Info
10	11.469000	192.168.10.13	192.168.10.11	ICMP	78	Echo (ping) reply id=0x35c6, seq=2/512, ttl=128 (reply in 9)
11	12.485000	192.168.10.11	192.168.10.13	ICMP	78	Echo (ping) request id=0x36c6, seq=3/768, ttl=128 (reply in 12)
12	12.547000	192.168.10.13	192.168.10.11	ICMP	78	Echo (ping) reply id=0x36c6, seq=3/768, ttl=128 (request in 11)
13	13.391000	HuaweiTechno_41:75::	Spanning-tree-(for-...	STP	119	MST. Root = 32768/0/4c:1f:cc:41:75:f2 Cost = 0 Port = 0x8018
14	13.579000	192.168.10.11	192.168.10.13	ICMP	78	Echo (ping) request id=0x38c6, seq=4/1024, ttl=128 (reply in 15)
15	13.641000	192.168.10.13	192.168.10.11	ICMP	78	Echo (ping) reply id=0x38c6, seq=4/1024, ttl=128 (request in 14)
16	14.672000	192.168.10.11	192.168.10.13	ICMP	78	Echo (ping) request id=0x39c6, seq=5/1280, ttl=128 (reply in 17)
17	14.719000	192.168.10.13	192.168.10.11	ICMP	78	Echo (ping) reply id=0x39c6, seq=5/1280, ttl=128 (request in 16)
18	15.532000	HuaweiTechno_41:75::	Spanning-tree-(for-...	STP	119	MST. Root = 32768/0/4c:1f:cc:41:75:f2 Cost = 0 Port = 0x8018
19	17.766000	HuaweiTechno_41:75::	Spanning-tree-(for-...	STP	119	MST. Root = 32768/0/4c:1f:cc:41:75:f2 Cost = 0 Port = 0x8018
20	28.032000	HuaweiTechno_41:75::	Spanning-tree-(for-...	STP	119	MST. Root = 32768/0/4c:1f:cc:41:75:f2 Cost = 0 Port = 0x8018
>	Frame 11:	78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface 0				0000 54 89 98 5b 30 ed 5d 49 98 c0 47 8a 01 00 00 0a T [0-T] G-----
>	Ethernet II, Src:	HuaweiTechno_c0:47:8a (54:89:98:c0:47:8a), Dst:				0010 08 05 40 00 00 3c c6 40 00 80 01 9f 21 cc 0a E-E<6 @---
>	802.1Q Virtual LAN,	PRI: 0, DEI: 0, ID: 10				0020 0a 0b 0e 00 00 00 00 4f b5 36 c6 00 03 98 98 O 6-
>	Internet Protocol Version 4,	Src: 192.168.10.11, Dst: 192.168.10.13				0030 0a 0b 0c 0d 0e 0f 11 12 13 14 15 16 17 18 19 "#\$%
>	Internet Control Message Protocol					0040 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27

(7) RTA 上接口 2 捕获到广播帧, 携带的 VLAN ID 为 10:

8 15.563000	HuaweiTechno_a9:33:..	Spanning-tree(for.. STP	119 MST. Root = 32768/0/4c:1f:cc:41:75:f2	Cost = 20000	Port = 0x8002
9 17.844000	HuaweiTechno_a9:33:..	Spanning-tree(for.. STP	119 MST. Root = 32768/0/4c:1f:cc:41:75:f2	Cost = 20000	Port = 0x8002
10 19.625000	HuaweiTechno_c0:47:8a	Broadcast ARP	64 who has 192.168.10.13? Tell 192.168.10.11		
11 20.104100	HuaweiTechno_a9:33:..	Spanning-tree(for.. STP	119 MST. Root = 32768/0/4c:1f:cc:41:75:f2	Cost = 20000	Port = 0x8002
12 22.328000	HuaweiTechno_a9:33:..	Spanning-tree(for.. STP	119 MST. Root = 32768/0/4c:1f:cc:41:75:f2	Cost = 20000	Port = 0x8002
13 24.516000	HuaweiTechno_a9:33:..	Spanning-tree(for.. STP	119 MST. Root = 32768/0/4c:1f:cc:41:75:f2	Cost = 20000	Port = 0x8002
14 26.766000	HuaweiTechno_a9:33:..	Spanning-tree(for.. STP	119 MST. Root = 32768/0/4c:1f:cc:41:75:f2	Cost = 20000	Port = 0x8002
15 28.969000	HuaweiTechno_a9:33:..	Spanning-tree(for.. STP	119 MST. Root = 32768/0/4c:1f:cc:41:75:f2	Cost = 20000	Port = 0x8002
16 31.203000	HuaweiTechno_a9:33:..	Spanning-tree(for.. STP	119 MST. Root = 32768/0/4c:1f:cc:41:75:f2	Cost = 20000	Port = 0x8002
17 33.407000	HuaweiTechno_a9:33:..	Spanning-tree(for.. STP	119 MST. Root = 32768/0/4c:1f:cc:41:75:f2	Cost = 20000	Port = 0x8002
<pre>> Frame 10: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface 0 Ethernet II, Src: HuaweiTechno_c0:47:8a (54:89:98:c0:47:8a), Dst: Broadcast (ff:ff:ff:ff:ff:ff) 802.3 Virtual LAN, PRI: 0, DEI: 0, ID: 10 Address Resolution Protocol (request)</pre>					
0000	ff ff ff ff ff ff	54 89 c0 47 8a 81 00 00 0a	T.....G.....		
0010	08 00 00 01 08 00 06 04	00 81 54 89 98 c0 47 8a	G.....T.....G.....		
0020	c0 a8 0a 00 ff ff ff ff	ff ff c0 a8 0a 00 00 00	G.....T.....G.....		
0030	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	T.....G.....		